Ctrl+Z Burnout

Academic Decision Simulator

1. Journey Maps

Samuel: returning working student, dyslexia

Goal: fit 2 evening courses around a 25 h/week job without a reading-heavy overload.

Stage	What Samuel does	Pain point	Where User Control and Agency helps
Start	Opens the simulator after a long gap from school	Fears he'll be overwhelmed by dense text and jargon	Onboarding offers a **display choice** (dyslexia-friendly font, spacing, read-aloud) before any content. He chooses the pace.
Ask	Enters constraints: max 6 credits, evenings only, 25 work hours, "avoid reading-heavy courses"	Free-text typing is slow and error-prone	The **constraints dashboard** has sliders, toggles and dropdowns. The AI echoes back "Here's what I understood" so he can correct it before generating.
Understand	Reads the draft schedule and its explanation	Long paragraphs, ambiguous terms (e.g. "co-requisite")	Plain-language summary first, details on demand. Each course shows a **workload bar** (reading hrs/week) instead of prose.

Stage	What Samuel does	Pain point	Where User Control and Agency helps
Act or fix	Sees a course with 9 h/week of reading. Rejects it, lowers his weekly-hours limit to 18 h, then clicks **Undo** after a regretted change	Fear that one wrong click ruins his plan	**Reject, Edit limit and Undo** are always visible. Undo restores the exact previous state, so mistakes cost nothing.
Finish	Saves a plan he trusts and exports it	Unsure the plan is "safe"	A final **review screen** lists his own limits next to the plan's totals. He confirms explicitly, and nothing is auto-saved.

Lucía: first-year commuter, ADHD

Goal: build a first-semester plan that works with a 90-minute commute.

Stage	What Lucía does	Pain point	Where User Control and Agency helps
Start	Opens the tool between classes	Decision paralysis, low tolerance for setup	A one-screen start with 3 questions. She can skip optional fields and return later.
Ask	Enters credit load, commute days, "no 8am classes"	Forgets constraints she hasn't stated (e.g. lab hours)	The dashboard shows **constraint chips** she can add or remove. The AI prompts about commonly missed constraints but never forces them.

Stage	What Lucía does	Pain point	Where User Control and Agency helps
Understand	Reviews 3 draft options	Too many options and too much text at once	**Capped at 3 drafts**, shown one at a time with a 3-line summary and a "why this fits you" tag.
Act or fix	Locks her "no 8am" rule, rejects a back-to-back day, and regenerates	Losing track of what she changed	A **change log** with Undo for each step. **Locked** constraints stay fixed across regenerations.
Finish	Picks one draft and gets a checklist of next steps	Drops off before completing registration steps	The system gives a short "what's next" list. She decides when and whether to proceed.

Nour: advanced international CS student

Goal: finish remaining CS requirements on time while meeting a full-time credit requirement.

Stage	What Nour does	Pain point	Where User Control and Agency helps
Start	Imports completed courses and the remaining requirement list	Distrusts AI that may miss prerequisites or program rules	A **"what this tool knows and doesn't"** panel states the data source (course catalog) and its limits.
Ask	Sets min 12 credits, a capstone-first priority, and max 2 heavy-workload courses	Needs to express hard rules vs. soft preferences	Each constraint is marked **hard** (never violated) or **soft** (preferred), and she controls which is which.

Stage	What Nour does	Pain point	Where User Control and Agency helps
Understand	Checks the prerequisite chain behind each suggestion	Opaque reasoning, can't verify claims	Every course shows its **source catalog entry** and prerequisite path. Uncertain items are flagged.
Act or fix	Rejects an elective that doesn't count for her track, edits her heavy-course limit, and flags a wrong prerequisite	AI errors are costly and hard to report	**Reject, Edit and "Report an error"** apply immediately and trigger a re-plan. Undo reverses any of them.
Finish	Compares two final plans and exports one	Wants a graduation-timeline check, not just a schedule	A **side-by-side compare** with the totals checked against her hard constraints. She approves, and the AI never auto-finalizes.

2.Measurable Objectives:

#	Type	Objective	Target	How measured
O1	AI quality	Prerequisite and co-requisite accuracy of generated plans	**≥ 95%** of course recommendations valid, with **0 courses missing from the catalog**	Run 200 generated plans against a hand-verified gold set built from the synthetic catalog
O2	UX	Time from first input to first usable draft	**Median ≤ 90 seconds** (no participant over 150 s)	Timed usability sessions, n = 9 (3 per persona)

#	Type	Objective	Target	How measured
O3	HAI: User Control and Agency	Share of users who successfully complete the control task (reject a course, edit a limit, undo) **unaided**	**≥ 85%** (≥ 8 of 9 participants), with Undo restoring the prior state **100%** of the time	Task-based usability test plus automated Undo state-restore tests
O4	Project success	Reduction in overloaded schedules (weekly workload above the user's own stated limit)	**≥ 40% relative reduction** vs. participants' unaided baseline, and **≤ 10%** of generated drafts overloaded	Compare baseline manual plans with simulator plans using the workload dataset

3.HAI Principle Ladder v1: User Control and Agency

Level	What it means in our product	Persona example	Linked objective	Pass criterion
1. Inform	The user always knows what the system understood, used and cannot do (constraint echo, data-source panel, status indicators)	Samuel confirms "Here's what I understood" before generating	**O2** (UX): clear, visible inputs support fast drafts	Median time to draft ≤ 90 s
2. Explain	Every recommendation shows why (source catalog entry, prerequisite path, workload breakdown) and flags uncertainty	Nour traces a prerequisite chain to its source	**O1** (AI quality): explanations must be verifiable against the catalog	Prerequisite accuracy ≥ 95%

Level	What it means in our product	Persona example	Linked objective	Pass criterion
3. Allow control	Users can reject, edit, lock and undo at any step, and nothing is final without their confirmation	Lucía locks "no 8am" and undoes a change	**O3** (HAI): control features must be discoverable and usable	≥ 85% unaided success, Undo 100% reliable
4. Enable feedback	Users can report errors and rate drafts, and the system re-plans using that feedback	Nour flags a wrong prerequisite and sees the re-plan	**O4** (project success): feedback loops reduce overloaded plans	Overloaded schedules down ≥ 40%

4.Tool Choices

Layer	Choice	Role
LLM	Hosted API (OpenAI or Gemini, a fast mid-tier model)	Generates explanations and draft proposals, using low temperature and structured JSON output
Orchestration	**LangChain**, using LangGraph for stateful flows	Chains retrieval → draft → validation → explanation. State checkpoints in the graph support **Undo**.
Frontend	**Streamlit**	Constraints dashboard (sliders, toggles, constraint chips), draft viewer, change log, and Undo/Reject/Lock buttons. st.session_state holds the history stack.

Layer	Choice	Role
Grounding	**RAG** over the synthetic course catalog and workload dataset, in a vector store (Chroma or FAISS) with embeddings	The LLM sees only retrieved catalog entries. Every claim cites its source record.
Validation	Deterministic rule checker (prerequisite graph, credit and workload calculator)	The LLM proposes and the checker verifies, which drives O1 and O4. Failures are fixed or flagged, never silently shown.

Accessibility note: dyslexia-friendly fonts, spacing and read-aloud will be built into Streamlit via custom CSS and a text-to-speech option. They are part of "Inform" and "Allow control", not extras.

Assumptions to confirm: the sample size (n = 9) is small, so results are directional and not statistically conclusive. The 40% overload reduction needs a baseline measured early. Credit minimums such as Nour's are treated as user-entered constraints, not as legal or visa advice.

Prompt Log:

#	Tool	Exact prompt	What it gave	What we changed and why
4	Claude	Act as a Lead UX Researcher and AI System Architect. I am developing the "Academic Decision Simulator" (Team: Ctrl+Z Burnout). The mandatory HAI principle is "User Control and Agency". We have 3 established personas: Samuel (returning working student with Dyslexia), Lucía (first-year commuter with ADHD), and Nour (advanced international CS student). Please generate our "Week 4:	Generated comprehensive Journey Maps for all 3 personas, 4 measurable objectives with exact targets, a 4-level HAI ladder mapped to objectives, and a specific tech stack (Streamlit,	**Change:** We reviewed and accepted the core logic, ensuring the tech stack aligned with our team's capabilities. We verified that the 90-second UX target and 95% accuracy targets are realistic for our testing phase. **Why:** To confirm the AI's proposed metrics are genuinely

#	Tool	Exact prompt	What it gave	What we changed and why
		Objectives and Ladder (Milestone 2)" deliverables based on the following strict constraints: 1. Journey Maps (One for each of the 3 personas): Map their journey through the simulator. For each, include: Start, Ask (inputting constraints), Understand the answer, Act or fix a problem (e.g., using "Undo", editing a limit, rejecting a course), and Finish. Clearly highlight their pain points and exactly where the "User Control and Agency" principle helps them. 2. Measurable Objectives (with exact target numbers): Provide exactly 4 measurable objectives: • 1 AI-quality metric (e.g., prerequisite accuracy). • 1 User-experience (UX) metric (e.g., time to generate a draft). • 1 metric directly tied to our HAI principle "User Control and Agency" (e.g., percentage of users successfully using the Undo/Edit features). • 1 project success metric (e.g., reduction in overloaded schedules). 1. HAI Principle Ladder v1: Map our principle across exactly 4 levels: 1. Inform, 2. Explain, 3.	LangChain, RAG).	measurable during our upcoming usability testing and to lock in the technical architecture for the next phase.

#	Tool	Exact prompt	What it gave	What we changed and why
		Allow control, 4. Enable feedback. Link each of these 4 levels directly to one of the measurable objectives defined above. 2. Tool Choices: Briefly specify the tech stack: A hosted LLM API (e.g., OpenAI or Gemini), a backend/workflow framework (LangChain), a frontend UI (Streamlit for the constraints dashboard), and how answers will be grounded (e.g., using RAG / Vector search on our synthetic course catalog and workload dataset).		